

Modeling User Requirements for Inclusive Pedestrian Routing with the PIIRRITE Ontology

Salomon van Ginneken^{1,*}, Nadia Yacoubi Ayadi², Stéphanie Jean-Daubias² and Gilles Gesquière¹

¹Université Lumière Lyon 2, CNRS, INSA Lyon, Université Claude Bernard Lyon 1, LIRIS, UMR5205, 69007 Lyon, France

²Université Claude Bernard Lyon 1, CNRS, INSA Lyon, LIRIS, UMR5205, 69622 Villeurbanne, France

Abstract

Inclusive pedestrian routing requires not only a semantic description of the traversed environment and its dynamic contextual properties, but also expressive models of users' requirements. Existing ontology-based user profiling approaches primarily organize user characteristics into taxonomies, providing limited support for representing conditional requirements that can be evaluated against changing environmental properties. Conversely, ontologies describing pedestrian environments focus mainly on spatial and contextual information, with little support for modeling the structure and semantics of user requirements. This paper extends the PIIRRITE ontology with two new modelets for representing user requirements, including the accessibility-related ones, in pedestrian routing. Our model captures user profiles as sets of conditional requirements, distinguishes requirement importance from flexibility, supports recursively composed requirements, and introduces a distance factor mechanism for proximity-sensitive constraints. We validate the ontology by instantiating a knowledge graph of the LyonTech La Doua campus with representative user personas, evaluating SPARQL queries implementing a set of competency questions, and computing an inadequacy score quantifying a route's suitability for a given user profile.

Ontology: <https://github.com/Alicante2/PIIRRITE-ontology/blob/master/PIIRRITE.owl>

Documentation: <https://aicante2.github.io/PIIRRITE-ontology/Documentation/index-en.html>

WebVOWL: <https://aicante2.github.io/PIIRRITE-ontology/webvowl/>

GitHub: <https://github.com/Alicante2/PIIRRITE-ontology>

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w3id IRI: <https://w3id.org/piirrite/core#>

Keywords

Inclusive-by-Design, Pedestrian itinerary, Ontology Engineering, Semantic Interoperability, User Profiling

1. Introduction

Pedestrian mobility poses major challenges in modern society and requires a *design for all* approach [1] capable of meeting everyone's diverse needs. Whether navigating a university campus, visiting a museum, or exploring a downtown area, pedestrians can greatly benefit from route-planning solutions that go beyond simply optimizing the shortest route, by taking into account the specific needs, characteristics, and preferences of each user. Such an inclusive system requires a thorough description of the traversed environment and its contextual properties — wheelchair accessibility, protection from adverse weather, elevator availability, noise reduction or proximity to particular points of interest such as rest areas or amenities.

Supporting such personalized, inclusive routing requires more than an accurate description of the environment: it requires a model of the user capable of expressing, with relevant granularity, the nature of her requirements. Not all requirements carry the same importance, nor should they be enforced with the same rigidity. A wheelchair user's need for step-free access constitutes a hard constraint, whereas a preference for quieter itineraries may only be a soft preference. Individual requirements can be further combined into complex conditions, e.g., *avoid stairs unless no elevator is available within 50 meters*, that most existing profiling models are not able to express. Reconciling this diversity of requirement types

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*Corresponding author.

✉ salomon.van-ginneken@univ-lyon2.fr (S. van Ginneken); nadia.yacoubi-ayadi@univ-lyon1.fr (N. Yacoubi Ayadi); stephanie.jean-daubias@univ-lyon1.fr (S. Jean-Daubias); gilles.gesquiere@univ-lyon2.fr (G. Gesquière)

🆔 0009-0009-0279-7192 (S. van Ginneken); 0000-0002-6132-8718 (N. Yacoubi Ayadi); 0000-0003-2209-6376 (S. Jean-Daubias); 0000-0001-7088-1067 (G. Gesquière)



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with heterogeneous, dynamic environmental data —whose contextual properties (weather, crowdedness, pollen levels) evolve over time— constitutes a central challenge for any inclusive routing system.

Ontology-based user profiling has been studied extensively [2, 3, 4]; however, existing models are typically organized as taxonomies of user dimensions (interests, knowledge, preferences) rather than as reusable patterns for expressing conditional requirements against observed and dynamic properties. Existing semantic models addressing itinerary modeling and route planning [5, 6, 7] focus primarily on the representation of spatial and environmental data, offering limited support for representing the structure, relative importance and flexibility of user requirements themselves. To the best of our knowledge, no existing ontology provides a reusable, domain-agnostic design pattern for expressing user profiles as sets of conditional, and possibly composed requirements that can be matched against dynamic contextual properties.

This paper addresses this gap. Building on PIIRRITE¹ (Pedestrian Inclusive indoor-outdoor ITinerariEs), whose first three modelets provide a semantic backbone for representing spatial entities and their time-dependent contextual properties, we introduce and detail new modelets dedicated to modeling user requirements in regard of inclusive pedestrian routing. Specifically, this paper makes the following contributions:

- We present the PIIRRITE schema and introduce a reusable Ontology Design Pattern (ODP) for representing user profiles as sets of conditional *requirements* over contextual properties. We distinguish the *importance* of a requirement and the *flexibility* with which it should be enforced.
- We extend this pattern to support *composed* requirements, expressed as recursively nested logical combinations of conditions, together with a *distance factor* mechanism for expressing proximity-dependent constraints (e.g., avoiding entities located within a given distance of the itinerary).
- We ground the design of these modelets in real-world requirements elicited from representative personas, developed in collaboration with a domain expert in occupational therapy and refined through public outreach workshops on inclusive routing. We demonstrate the applicability of the resulting model through SPARQL queries addressing the competency questions, and through a use case computing a user-specific *inadequacy score* for a navigable path, directly usable as edge weights for a shortest-path algorithm.

The PIIRRITE ontology complies with the FAIR principles: it is *findable* through a persistent DOI (10.5281/zenodo.21496238) and a stable IRI (w3id.org/piirrite/core); it is *accessible* via a public GitHub repository and freely available online documentation; and it is *interoperable* and *reusable* thanks to its standard OWL format, its WebVOWL visualization and its explicit CC BY-NC-SA 4.0 license specifying the reuse conditions.

This paper is structured as follows. Section 2 introduces two of the representative user personas whose requirements guided our modeling decisions. A set of competency questions is also introduced to specify the conceptual requirements that the PIIRRITE ontology must satisfy. Section 3 presents the PIIRRITE ontology by providing firstly a short summary of how the first three modelets model the contextual properties of the traversed environment. We then focus on the most recently designed parts of the PIIRRITE ontology, which cover complex, composed user requirements. In Section 4 we argue that in its current state, the PIIRRITE ontology is able to cover most of the needs initially identified. Related works are presented in Section 5. Finally, Section 6 discusses how some details remain untouched and proposes ways of covering them.

2. Motivation and Competency Questions

2.1. From shortest paths to convenient paths

As a representative real-world environment, we take the LyonTech La Doua campus as a navigable environment. A subpart of this campus is illustrated in Figure 1, which contrasts two itineraries connecting points A and B. The shortest route (shown in blue) leaves the starting building directly, follows a tram line and arrives from the west. Another possible route (orange line), which may prove more convenient depending on the user’s preferences, passes through several indoor areas and arrives

¹<https://aiicante2.github.io/PIIRRITE-ontology/Documentation/index-en.html>

from the east. Although shorter in distance, the blue itinerary exposes the user to outdoor weather, tramway noise, and a potentially non-accessible entrance, which may pose difficulties to some users. The orange itinerary maximizes indoor traversal, avoids noise, and reaches point *B* through a more accessible side, which are environmental characteristics that may be looked after by some user. Identifying such an itinerary therefore requires accounting for contextual characteristics as well as user preferences and constraints, including sensory factors (e.g., noise level), physical accessibility (e.g., entrance type, vertical transitions), and dynamic conditions (e.g., weather, lighting).

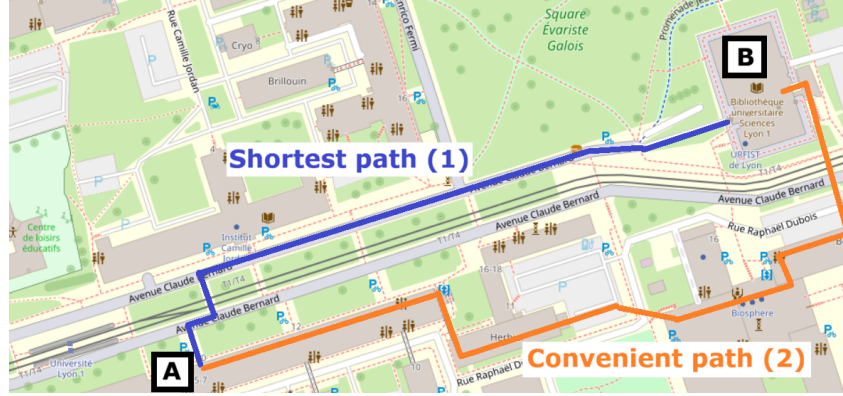


Figure 1: Shortest path (blue) vs. convenient path (orange) on the LyonTech La Doua campus.

2.2. User Personas

In this section, we describe how we made sure to build an *inclusive-by-design* ontology. To this end, we collaborated with a domain expert with a background in occupational therapy, who contributed to the elaboration of a reference vocabulary cataloging environmental characteristics considered as *potentially relevant* to users in pedestrian mobility. Building on this work, we derived several personas, two of which are presented below.

The first persona, **Anna**, is a first-year undergraduate student in Computer Science diagnosed with autism spectrum disorder. She feels uncomfortable in crowded or noisy places and needs opportunities to rest in quiet spaces. Based on Anna’s profile, we derived the following requirements:

- R_1 The routing system should characterize spatial areas through relevant contextual properties (e.g. noise level, crowdedness).
- R_2 The system should identify and help to avoid high-traffic areas during peak hours.
- R_3 The system should be able to locate rest areas along or near to potential routes.

The second persona, **Sacha**, is a master’s student in Sports Science who prefers physically engaging routes (stairs over elevators, outdoor green spaces) and occasionally needs quick access to specific amenities. Her profile motivates the following requirements:

- R_4 The system should distinguish between vertical transition types (stairs, elevators, escalators) to support preference and capability-aware routing.
- R_5 The system should locate the nearest instance of a desired amenity type (vending machine, toilets, reception board) to the current proposed itinerary.
- R_6 The system should characterize spaces based on their aesthetic or comfort-related qualities to enable experience-aware routing.

2.3. Competency Questions

Following the requirements presented above, we defined a set of competency questions (CQs) that both scope the domain and serve to validate the resulting ontology. To organize them, we rely on the typology proposed by Keet and Khan [8]. The entire list of CQs is provided in our GitHub².

²<https://github.com/Alicante2/PIIRRITE-ontology>

2.3.1. Scoping Competency Questions

Scoping CQs delimit the subject domain by specifying which entities, and which of their characteristics, fall within the intended coverage of the ontology [8]. Beyond demarcating this scope, they support the modeling process by making explicit the requirements the system is expected to satisfy.

SCQ₁. What types of spatial entities constitute a navigable environment? The expected answer should comprise different spatial entity types: points, representing dimensionless spatial entities (e.g., a road intersection, a vending machine); polylines, generalizing linear segments; areas, possibly approximated by polygons (e.g., a park, the noise-affected zone surrounding renovation works); and navigable paths, specializing polylines (e.g., a sidewalk, a gravel path).

SCQ₆. How can the set of contextual properties relevant to Anna's requirements be represented? This CQ addresses the model's dual ability to structure the contextual properties characterizing elements of the navigable environment, and to describe a user's requirements in terms of the contextual properties they constrain. Answering it requires linking the user to her set of individual requirements and, for each requirement, identifying the contextual property it targets.

SCQ₇. How can we distinguish, for a given requirement, between (i) its importance and (ii) its flexibility? This distinction emerged from feedback collected during workshops aimed at introducing a non-specialist audience to the challenges of inclusive routing. The discussions revealed the need to associate each user requirement with two complementary, user-defined scores. The first score captures the importance of a requirement to the user, reflecting the level of inconvenience experienced if the requirement is not satisfied. The second score is used during the matching process between the user's expected value and the actual value of the corresponding property. It quantifies the impact of any discrepancy between these values on the overall suitability of the entity.

2.3.2. Relationship Competency Questions

Relationship CQs assess how entities relate to one another, and with what cardinality, arity, or relation properties [8].

RCQ₁. Is every piirrite:NavigablePath associated with at least one piirrite:PointOfInterest? The expected answer is No. The ontology does not impose a minimum cardinality constraint on the piirrite:passesBy relation for instances of piirrite:NavigablePath instances. This design choice reflects the fact that some paths are intended only for navigational purposes and do not necessarily provide access to any piirrite:PointOfInterest.

RCQ₆. How many piirrite:Condition instances are required to define a piirrite:Requirement? The expected answer is at least one. A requirement without an associated condition would be semantically incomplete, as it would fail to define any criterion for evaluating the suitability of an environmental feature with respect to the user's needs.

2.3.3. Validation Competency Questions

Validation CQs check that the ontology and the data it organizes can answer the queries expected of the running system [8]. For the profiling modelets, they exercise the matching of user requirements against the possibly time-varying environmental properties.

VCQ₄. Where is the nearest vending machine to a bicycle park with a capacity of at least 20? In addition to geometry-based distance calculation, this question ensures that the PIIRRITE ontology should enable (i) retrieving of spatial entities of a given type (vending machines, bicycle parking lots) and (ii) their filtering based on a quantitative contextual sub-property (parking capacity).

VCQ₉. What is the subset of Anna's requirements that involve at least one distance-related factor? This competency question assesses whether PIIRRITE can represent the association between users, their requirements, and the conditions that characterize these requirements, including the relationships among such conditions. By providing these semantics, PIIRRITE supports the modeling of complex user

needs with respect to the traversed environment and establishes a foundation for the development of comprehensive and personalized *inclusive itinerary recommendation systems*.

$VCCQ_{10}$. *What is the maximum degree of recursion achieved by any composed condition?* This CQ (i) checks that no absurdly-deep composed condition exists in the instantiated KG, (ii) goes through every composed condition and ensures that each one is well-built and (iii) give an idea of how complex the user requirements get in the considered instantiation of PIIRRITE.

3. The PIIRRITE Ontology

Given the concrete requirements and competency questions (CQs) presented in Section 2, we designed a semantic model aimed at structuring and integrating spatio-contextual information originating from heterogeneous data sources. The model was conceived to capture the multiple dimensions involved in inclusive pedestrian mobility, including environmental characteristics, accessibility-related features, and user-specific preferences and constraints.

The PIIRRITE ontology has been designed based on the principles of the SAMOD methodology, which gives guidelines for an incremental and modellet-driven conception [9]. In Section 3.1, we introduce the first three modellets of the PIIRRITE ontology. Rather than designing a single monolithic ontology, we adopt a pattern-based approach favoring the incremental composition of small, self-contained modules, each validated against its own subset of CQs before being integrated into a coherent whole.

3.1. Spatio-temporal and contextual modeling for inclusive itineraries

The core schema of the PIIRRITE ontology is depicted in Figure 2. The schema is organized into three complementary modellets: (1) the representation of navigable entities and their geometries, extending GeoSPARQL[10] and Simple Features; (2) the temporal validity of environmental information, based on OWL-Time[11]; and (3) the description of environmental features and their properties, reusing SAREF[12], QUDT[13], and SKOS[14]. More details are provided in following.

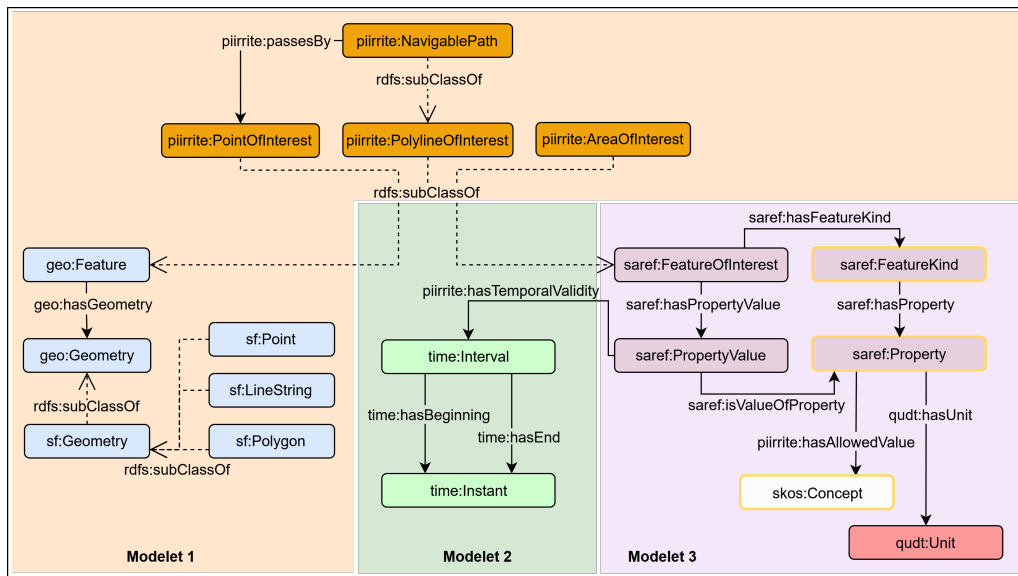


Figure 2: The core schema of the PIIRRITE ontology capturing spatial, contextual, and temporal dimensions of navigable environments.

Modellet 1: Spatial Representation of Entities. The first modellet addresses the fundamental requirement of any routing system: representing the spatial entities (e.g. points of interest, navigable polylines, areas) that constitute the physical environment and the topological relations between them. All three

interest classes are defined as subclasses of `geo:Feature`, thereby inheriting the full spatial representation and querying mechanisms of GeoSPARQL. The `piirrite:NavigablePath` class is defined as a specialization of `piirrite:PolylineOfInterest` and is related by the `piirrite:passesBy` object property to a set of `piirrite:PointOfInterest` instances. By leveraging GeoSPARQL vocabulary, this layer enables basic spatial filtering and geometric reasoning over the navigable environment.

Modelet 2: Contextual Characterization. The second modelet of the PIIRRITE ontology introduces the semantic layer needed to characterize environmental properties (e.g. noise levels, crowdedness, pollen presence) that impact the user experience. This modelet aligns with the SAREF *Ontology Design Pattern* for properties [12] and employs SKOS controlled vocabularies [14] to formally define properties and features. To address this requirement, Modelet 2 extends the four PIIRRITE classes introduced in Modelet 1 as subclasses of both `geo:Feature` (inherited from Modelet 1) and `saref:FeatureOfInterest`, establishing a dual alignment that integrates spatial and contextual description within a single class hierarchy.

Modelet 3: Temporal Validity of Contextual Properties. Contextual properties relevant to inclusive itineraries are often time-dependent (e.g. weather conditions, daily pollen concentrations). Modelet 3 extends the PIIRRITE ontology by providing a design choice for the temporal validity of these properties values. By reusing the OWL-Time ontology, this layer associates `saref:PropertyValue` instances with `time:Interval` entities.

3.2. User profiling and requirements modeling

Existing approaches to user preference modeling generally do not distinguish hard constraints (e.g., a required wheelchair-accessible path) from soft preferences (e.g., a preference for quieter routes), nor do they express how strictly the latter should be enforced. The fourth modelet of the PIIRRITE addresses this gap by introducing a pattern for representing user profiles as sets of conditional *requirements* over contextual properties of the environment. It is illustrated in the right part of Figure 3.

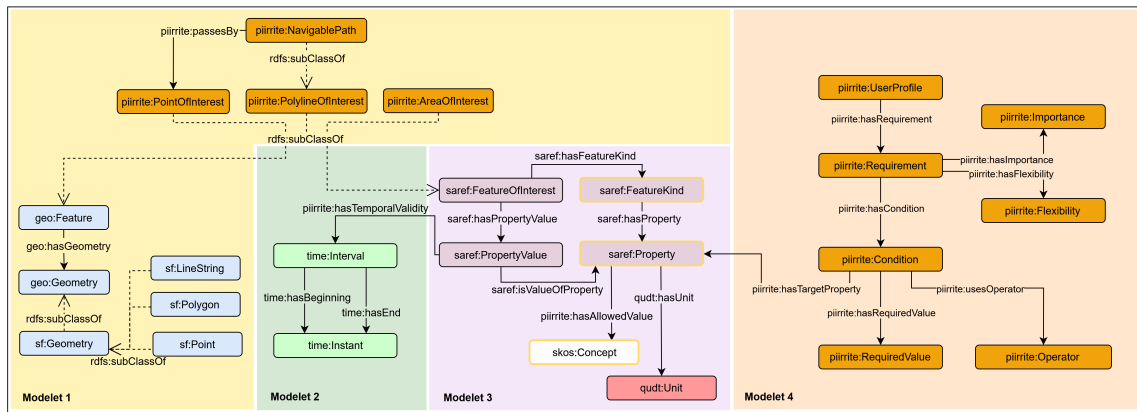


Figure 3: The 4th modelet introduces a design pattern for modeling users profiles based on their requirements.

3.2.1. User profiling through requirements

The `piirrite:UserProfile` class acts as a container for all the requirements held by a given user (e.g., :Anna), linked to each of them via the `piirrite:hasRequirement` object property. Each `piirrite:Requirement` instance carries a `piirrite:Importance` and a `piirrite:Flexibility`, both value wrappers expressing user-defined coefficients in $[0,1]$ (a detailed example is presented in Section 4.2), and is associated with one or more `piirrite:Condition` instances. Third, each `piirrite:Condition` specifies how a requirement is to be evaluated against the environment, through three object properties:

- `piirrite:hasTargetProperty`, linking the condition to a `saref:Property` that identifies the contextual property being constrained (e.g., `pcv:Noise`, `pcv:GroundWear`);
- `piirrite:hasRequiredValue`, linking the condition to a `piirrite:RequiredValue` that specifies the value the property must take for the condition to be satisfied (e.g., 60, ["low", "none"]);
- `piirrite:hasOperator`, linking the condition to a `piirrite:Operator` used to compare the required value against the environment's actual property value (e.g., `piirrite:LESS_THAN`, `piirrite:IN`). By convention, the required value is treated as the left operand and the actual value as the right operand of this comparison.

Through its 4th modellet, PIIRRITE can represent user requirements and allow a robust and easy matching against the environmental context. However, it cannot grasp complex, composed user requirements like "I want to avoid tramway noise or construction work noise" or "I want to avoid being 50m from a crowded place". The 5th and last-to-date modellet of PIIRRITE allows for the modeling of such complex user requirements.

3.2.2. Complex user requirements modeling

The first contribution of Modellet 5 is the decomposition of the `piirrite:Condition` class into two subclasses: `piirrite:AtomicCondition` and `piirrite:ComposedCondition`. The former is structurally equivalent to the `piirrite:Condition` class introduced in Modellet 4: it carries a target property, a required value, and a comparison operator. The latter carries none of these relations; its role is instead to compose several conditions by treating them as operands within a logical expression, using the logical operators. In such a composed expression, the operands are themselves `piirrite:Condition` instances—either atomic or composed—thereby allowing for recursive composition. For instance, a composed condition may express that *path p satisfies the wheelchair-accessible requirement if and only if the ground nature of p is smooth AND (the slope of p does not exceed 6° OR p is an elevator)*.

This modellet also introduces the class `piirrite:DistanceFactor`, which enables the definition of spatial constraints based on proximity. A `piirrite:DistanceFactor` is associated with an atomic condition, making the evaluation of that condition dependent on the distance between the spatial entity under consideration and other surrounding entities. This mechanism allows the representation of requirements such as “*Anna wishes to avoid passing within 50 m of pollinic plants.*” Each atomic condition may be associated with at most one distance factor:

- If no distance factor is specified, the condition is evaluated directly on the spatial entity under consideration.
- If a distance factor is associated with the condition, the evaluation is extended to all spatial entities located within a distance satisfying the specified constraint. The admissible distance is determined using the `piirrite:RequiredValue` together with the `piirrite:ComparisonOperator` associated with the distance factor. This combination enables the specification of distance-based constraints such as “less than 50 m” or “exactly 30 m”.

Finally, the 5th modellet provides a domain-agnostic Ontology Design Pattern (see the light blue cases of Figure 4): the Requirement–Condition structure can characterize any user preference expressed as a comparison-based constraint over an externally observed property, making it reusable in other contexts where user profiles must be reconciled against dynamic and heterogeneous data.

4. Validation

In this section, we demonstrate the practical applicability of the PIIRRITE ontology. More details on the Knowledge Graph used in this validation are given in [15]. First, we explain how we implemented and executed SPARQL queries that directly address the CQ presented in Section 2.3. Due to space constraints, only three Validating CQs are presented in this section. All implemented CQ are available in our GitHub repos-

3. The diverse personas we created and used during the science outreach workshops we have been conducting for the past few years

The query presented in Listing 1 retrieves the nearest vending machine to any bicycle parking whose capacity exceeds a threshold by joining two `piirrite:SpatialPoint` instances *via* `geof:distance`. This nested constraint is expressible because PIIRRITE models attributes as `saref:Property` subclasses, supporting both type-based and value-based filtering. This query returns 1 result in 110–150ms.

Listing 1: SPARQL query implementing VCQ_4 .

```

SELECT ?vendingMachine ?vmGeom (MIN(?distance) AS ?minDistance)
WHERE {
  ?vendingMachine a piirrite:PointOfInterest ;
    geo:hasGeometry ?vmGeomNode .
  ?vmGeomNode geo:asWKT ?vmGeom .
  ?vendingMachine saref:hasPropertyValue ?pvVending .
  ?pvVending saref:isValueOfProperty pcv:Amenity ;
    saref:hasValue "vending_machine" .
  ?bicycleParking a piirrite:PointOfInterest ;
    saref:hasFeatureKind pcv:BicycleParking ;
    geo:hasGeometry ?bpGeomNode ;
    saref:hasPropertyValue ?pvCapacity .
  ?bpGeomNode geo:asWKT ?bpGeom .
  ?pvCapacity saref:isValueOfProperty pcv:P_Capacity ;
    saref:hasValue ?capacityValue .
  FILTER(xsd:integer(?capacityValue) >= 20)
  BIND(geof:distance(?vmGeom, ?bpGeom, uom:metre) AS ?distance)
}
GROUP BY ?vendingMachine ?vmGeom
ORDER BY ?minDistance
LIMIT 1

```

The implementation shown in Listing 2 illustrates how to identify, among a user’s requirements, those that rely on a distance factor, a client application can flag them for dedicated preprocessing, correctly evaluate them even when nested within composed requirements, and account for environmental data that influence the user’s itinerary only indirectly, i.e., through their proximity rather than their presence on the path itself. This query returns 1 result in 15–25ms.

Listing 2: SPARQL query implementing VCQ_9 .

```

SELECT DISTINCT ?requirement
WHERE {
  :Anna a piirrite:UserProfile ;
    piirrite:hasRequirement ?requirement .
  ?requirement piirrite:hasCondition ?condition .
  # Recursive search of atomic conditions inside composed conditions
  ?condition (piirrite:hasCondition|piirrite:hasOperand)* ?atomicCondition .
  ?atomicCondition a piirrite:AtomicCondition ;
    piirrite:hasDistanceFactor ?distanceFactor .
  ?distanceFactor a piirrite:DistanceFactor .
}

```

The implementation shown in Listing 3 goes through every `piirrite:UserProfile` and looks for the deepest `piirrite:ComposedCondition`. For instance, *A and B* has a recursion degree of 1 while the one of *not-(A or B) and C* is 3. This query returns 1 result in 20–30ms.

Listing 3: SPARQL query implementing VCQ_{10} .

```

SELECT (MAX(?nbConditions) - 1 AS ?maxRecursionDegree)
WHERE {
  {
    SELECT ?rootCondition (COUNT(DISTINCT ?anyCondition) AS ?nbConditions)
    WHERE {
      ?userProfile a piirrite:UserProfile ;
        piirrite:hasRequirement ?requirement .
      ?requirement piirrite:hasCondition ?rootCondition .
      ?rootCondition (piirrite:hasOperand)* ?anyCondition .
    }
    GROUP BY ?rootCondition
  }
}

```

4.2. Use case example: attributing an inadequacy score to a navigable path

To illustrate the use of the PIIRRITE ontology's 4th and 5th modelets, we present a short use case in which we assign an inadequacy score to a navigable path, given its contextual property values and two user requirements. We consider inadequacy rather than adequacy since *weighted shortest path* algorithms minimize traversed path weights rather than maximizing them. Let:

- *Anna* be a user, modeled as a `piirrite:UserProfile` that has two `piirrite:Requirement`:
 - $R_n = 50$: her itinerary should not exceed a noise level of 50 dB. Anna specified an importance of 0.75 and a flexibility of 0.25. It features a `piirrite:Condition` targeting `pcv:Noise` with the required value 50 and the operator `piirrite:LESSER_THAN`.
 - $R_t = 24$: her itinerary should feature a temperature of 24 °C. Anna specified an importance of 0.3 and a flexibility of 0.8. It features a `piirrite:Condition` targeting `pcv:Temperature` with the required value 24 and the operator `piirrite:EQUALS`.
- P be a path, modeled as a `piirrite:NavigablePath` characterized by two `saref:PropertyValue`:
 - $P_n = 57.3$ be the noise level (in dB) along P , a `saref:PropertyValue` of `pcv:Noise`
 - $P_t = 28.9$ be the temperature (in °C) along P , a `saref:PropertyValue` of `pcv:Temperature`

Before presenting the inadequacy functions themselves, we clarify how the values they operate on are obtained from the knowledge graph. For each `piirrite:Requirement` R_i , a SPARQL query, structurally similar to those presented in Section 4.1, jointly retrieves the Importance value i_i , the Flexibility value f_i , the target property t_i referenced by the associated `piirrite:Condition`, together with its required value and comparison operator. The observed value P_{t_i} of t_i along the candidate path P is retrieved analogously, from the `saref:PropertyValue` instances attached to P . Computing the inadequacy score then reduces to applying, for each R_i , a function $\bar{A}_i$ to the triple (R_i, f_i, P_{t_i}) retrieved in this way, provided $\bar{A}_i$ satisfies four properties: (i) it takes values in $[0, 1]$; (ii) it equals 0 when the condition of R_i is met; (iii) it increases as the condition of R_i becomes less satisfied; and (iv) it increases faster as the flexibility f_i decreases.

Figure 5 was generated using the inadequacy functions below; as expected, higher flexibility values yield lower inadequacy scores for both noise and temperature. Rather than mandating a single inadequacy function, our approach leaves this choice open to the consuming application, provided the selected function satisfies properties (i)–(iv) above – property (iv) in particular ensuring that a low Flexibility value amplifies the penalty incurred by an unmet requirement, independently of the implementation chosen. It illustrates two such admissible function, a smoothstep function for noise and a linear function for temperature. They were chosen arbitrarily and adjusted using numerical parameters to polish their shape to show that distinct functional forms, provided they satisfy (i)–(iv), remain mutually comparable and can be combined within a single weighted sum.

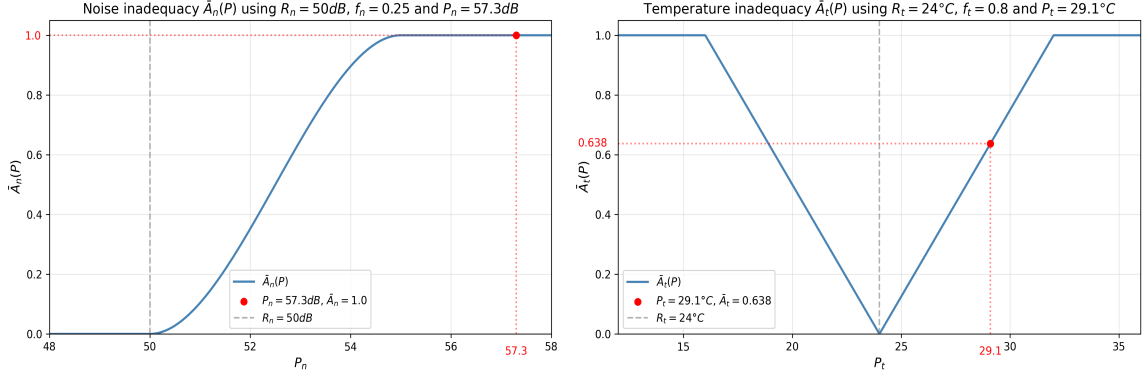


Figure 5: Noise and temperature inadequacies, computed from the given *Anna*'s profile and P parameters.

$$\bar{A}_n(P) = \text{smoothstep}(R_n, R_n + 20f_n, P) \quad \bar{A}_t(P) = \begin{cases} 1 & \text{if } P_t < R_t - 10f_t \\ \frac{P_t - R_t}{10f_t} & \text{if } P_t \in [R_t - 10f_t, R_t] \\ \frac{R_t - P_t}{10f_t} & \text{if } P_t \in [R_t, R_t + 10f_t] \\ 1 & \text{if } P_t > R_t + 10f_t \end{cases}$$

Then, the inadequacies are weighted by their respective importance and summed to yield the path's total inadequacy.

$$\begin{aligned} \bar{A}_P &= \sum_i \bar{A}_i(P) \cdot i_i \\ &= \bar{A}_n(P) \cdot i_n + \bar{A}_t(P) \cdot i_t \\ &= 1 \times 0.75 + 0.638 \times 0.3 \\ &= \mathbf{0.941} \end{aligned}$$

Repeating these operations over all navigable paths yields the data required to model a graph whose edges are the candidate navigable paths, weighted accordingly to the user requirements.

5. Related Work

Several works have shown the relevance of relying on semantic web technologies and linked data principles to allow semantic interoperability in the representation of spatio-temporal entities and their contextual properties [16, 17]. Building on this foundation, The research work presented in [18] proposes a modular meteorological ontology that combines modular ontologies such as SOSA/SSN [19], OWL-Time [11], QUDT, and GeoSPARQL [10] to publish meteorological Linked Data from the french national provider. Nativi *et al.* [20] further argue that environmental models are inherently constrained by the heterogeneity of the formats they rely upon (NetCDF grids, OGC coverages, sensor streams), positioning semantic technologies as the most principled layer for their integration. This claim finds empirical support in [7], where sensor feeds, raster data, OSM vectors, and weather data are semantically integrated within a unified smart city platform. At a larger scale, Zhu *et al.* [21] introduce a geospatial knowledge graph that reuses GeoSPARQL, SOSA/SSN, and OWL-Time to integrate more than 30 heterogeneous contextual assets, thereby enabling cross-domain spatial reasoning and rapid, geo-enriched context awareness. Novack *et al.* [22] present a pedestrian routing system built entirely on OpenStreetMap data, in which users interactively set, through discretized slide-bars, the relative importance of street length against three additional factors: greenness, sociability, and quietness; each directly quantified from OSM tags. While this system demonstrates the feasibility of multi-criteria, user-weighted pedestrian routing purely from crowd-sourced data, the weights themselves remain opaque numerical sliders external to the underlying

OSM data model, PIIRRITE's Requirement distinguishes between the importance a user assigns to a criterion and the flexibility with deviations from that criterion should be tolerated. Furthermore, the three factors considered are fixed by design and cannot be recursively combined into composed conditions, nor evaluated against contextual properties beyond what is statically encoded in OSM tags.

On another side, research on modeling user profiles with semantic web technologies has predominantly emphasized ontology-driven approaches for aggregating heterogeneous, distributed data into coherent, machine-readable representations. In the context of Information Retrieval (IR), Daoud *et al.* [2] propose a session-based personalized IR system that integrates a semantic user profile to better capture users' search interests over time. Hypergraph-based models have similarly been proposed to merge partial user profiles across domains, enriching them through semantic matching techniques [3]. More recently, the POAS framework illustrates how linked data standards and ontology engineering can be jointly leveraged to reconstruct rich academic user profiles from scattered online sources, although its scope remains limited to scholarly identity [4]. To the best of our knowledge, however, no consensual standard has yet emerged for the semantic modeling of user *requirements* specifically, as opposed to user profiles or interests more broadly. This gap constitutes a shortcoming that Modelets 4 and 5 of PIIRRITE aim to address, in particular through the design pattern for user requirements presented in sections 3.2.1 and 3.2.2.

6. Conclusion and Future works

This paper introduces PIIRRITE, an ontology providing a semantic foundation for (i) integrating heterogeneous spatio-contextual data in support of inclusive pedestrian routing, and (ii) representing complex user requirements to be matched against the contextual properties of the environment for the generation of inclusive itineraries. While Modelets 1 through 3 of PIIRRITE support the modeling of spatial entities and their time-dependent contextual properties, this paper focused on new Modelets 4 and 5, addressing respectively the modeling of basic and complex user requirements for inclusive pedestrian mobility. While the current version of the PIIRRITE ontology covers the core requirements identified in our motivating scenario; several directions remain open for future work.

A first direction focuses on data reliability. In a real world settings, PIIRRITE is expected to integrate data from heterogeneous sources, including community-contributed datasets (e.g., OpenStreetMap), institutional records (campus facilities, accessibility audits), sensor data streams (weather, noise, air quality), and user-generated feedback (e.g., obstacle reports). Such data sources are likely to vary substantially in completeness, update frequency, and trustworthiness; treating them as uniformly reliable risks yielding sub-optimal, or even unsafe, routing recommendations. Future modelets should therefore extend PIIRRITE with provenance and data quality metadata, for instance by aligning with the W3C Data Quality Vocabulary (DQV) [23].

Finally, we intend to move toward a full, end-to-end *proof of concept* by generating inclusive itineraries for evaluation by users. We plan to carry out this process in three stages:

1. Enriching the knowledge graph presented in Section 4.1, with additional data sources such as weather forecasts and historical records, noise measurements, air pollution levels, sun exposure, and crowdedness.
2. Integrating the resulting knowledge graph into either a purpose-built user interface or an existing, usable application. We plan to pursue the latter by embedding a PIIRRITE-based workflow into OPALE, a multi-service application developed at Université Lyon 1 [24].
3. Conducting standardized, reproducible user evaluations to assess the practical benefit of the generated itineraries.

Declaration on Generative AI

During the preparation of this work, the author(s) used the Claude Sonnet and GPT-5.4 models in order to: Check grammar and spelling. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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